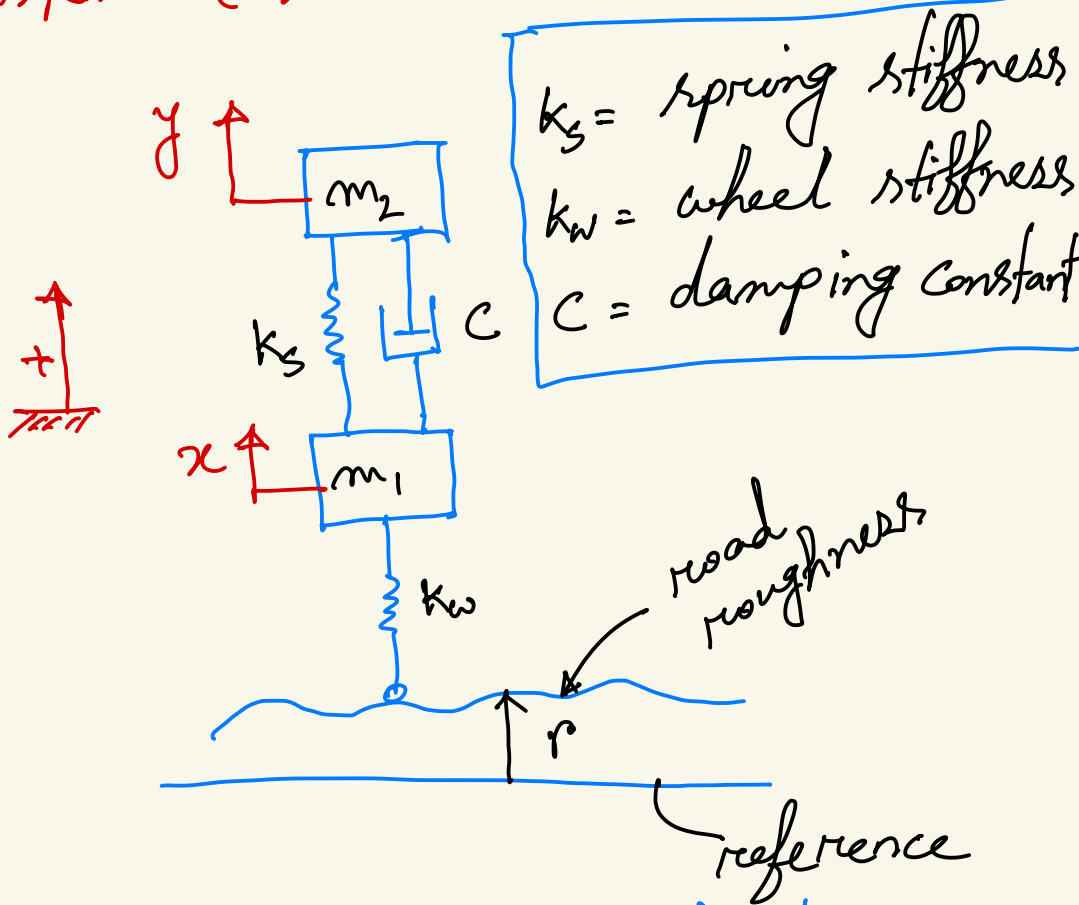




Modeling of a vehicle suspension system: (quarter car model)



m_1 = mass of the wheel

m_2 = mass of the car body

let's consider the freebody diagram under the displacement x and y .

$$c(\dot{x} - \dot{y}) \quad k_s(x - y)$$



$$k_w(x - r)$$



$$k_s(y - x) \quad c(\dot{y} - \dot{x})$$

Force balance for mass m_1

$$+\uparrow \sum F = m_1 \ddot{x}$$

$$\Rightarrow -k_s(x - y) - c(\dot{x} - \dot{y}) - k_w(x - r) = m_1 \ddot{x}$$

$$\Rightarrow -\frac{(k_s + k_w)}{m_1} x + \frac{k_s}{m_1} y - \frac{c}{m_1} \dot{x} + \frac{c}{m_1} \dot{y} + \frac{k_w}{m_1} r = \ddot{x}$$

----- (1)

force balance for mass m_2

$$\uparrow \sum F = m_2 \ddot{y}$$

$$\Rightarrow -k_s(y-x) - c(\dot{y}-\dot{x}) = m_2 \ddot{y}$$

$$\Rightarrow \frac{k_s}{m_2} x - \frac{k_s}{m_2} y + \frac{c}{m_2} \dot{x} - \frac{c}{m_2} \dot{y} = \ddot{y} \quad \text{--- (2)}$$

In order to formulate a state-space model, we need to perform change of variable, so that we can write in the following form:

$$\left. \begin{aligned} \dot{x} &= Ax + Bu \\ y &= Cx + Du \end{aligned} \right\} \text{state space form}$$

$$\begin{array}{l}
 x_1 = x \\
 x_2 = \dot{x} = \dot{x}_1
 \end{array}
 \left\{
 \begin{array}{l}
 \dot{x}_1 = \dot{x} = x_2 \\
 \dot{x}_2 = \ddot{x} = \text{equation (1)}
 \end{array}
 \right.$$

$$\ddot{x} = -\frac{(k_s + k_w)}{m_1} x + \frac{k_s}{m_1} y - \frac{c}{m_1} \dot{x} + \frac{c}{m_1} \dot{y} + \frac{k_w}{m_1} r$$

$$\begin{aligned}
 \dot{x}_2 = \ddot{x} = & -\left(\frac{k_s + k_w}{m_1}\right) x_1 + \frac{k_s}{m_1} x_3 \\
 & - \frac{c}{m_1} x_2 + \frac{c}{m_1} x_4 + \frac{k_w}{m_1} r
 \end{aligned}$$

$$\begin{array}{l}
 x_3 = y \\
 x_4 = \dot{y}
 \end{array}
 \left\{
 \begin{array}{l}
 \dot{x}_3 = \dot{y} = x_4 \\
 \dot{x}_4 = \ddot{y} = \text{equation 2} \\
 \dot{x}_4 = \ddot{y} = \frac{k_s}{m_2} x - \frac{k_s}{m_2} y + \frac{c}{m_2} \dot{x} - \frac{c}{m_2} \dot{y}
 \end{array}
 \right.$$

$$= \frac{k_s}{m_2} x_1 - \frac{k_s}{m_2} x_3$$

$$+ \frac{c}{m_2} x_2 - \frac{c}{m_2} x_4$$

With this modification, we can obtain A, B, C, and D matrix.

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -\frac{(k_s + k_w)}{m_1} & -\frac{c}{m_1} & \frac{k_s}{m_1} & \frac{c}{m_1} \\ 0 & 0 & 0 & 1 \\ \frac{k_s}{m_2} & \frac{c}{m_2} & -\frac{k_s}{m_2} & -\frac{c}{m_2} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{k_w}{m_1} \\ 0 \\ 0 \end{bmatrix} r(t)$$

$$y = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$

$$\text{If } m_1 = 150, \quad m_2 = 1500$$

$$k_s = 48000$$

$$k_w = 200000$$

A simulation with the above parameter value will result in the following figure.

